

SNEAKY SENSORS

Know What You Did Today!

Measuring, monitoring and controlling anything seem feasible with the current generation of wireless sensors that can match up to any of man's five senses. But, our exclusive sixth sense rings an alarming gong—will not such widespread use of sensors lead to security and privacy issues?



Freescale sensors

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Most of man's senses were at a time constrained by distance. He could see, hear, feel, taste or smell only what was near him. However, sensors came and changed all that. These little electronic nodes can act as eyes, ears, skin, nose or taste buds. They can be placed near or far, and even in environments that man cannot stand—biting cold, scorching heat, dangerous cliffs, toxic chemical vats and so on. And they do their owner's bid—measuring pressure (pressure sensors), temperature (heat sensors), vibration (vibration sensors), speed (motion sensors), brightness (light sensors), etc. What is more, they dutifully send this information to a computer or some other device that collates, inter-

prets and acts on the information sent by several sensors.

Sensors have not only made measurement and monitoring easier but also increased our control over processes, especially critical ones. This has made them imperative in this age and time when quality control and compliance with other regulations are very important. With sensors, the meteorology department can collect detailed information about weather conditions anywhere in the world, estate owners can watch the movement across their fences, process firms can monitor every minute detail of the environment and the assembly line that is required to ensure perfect execution, logistics firms can track their fleet and forest officers can monitor the movement of rare animals. In these and so many other ways, they let us watch the required details keenly and act immediately.

Endless uses, thanks to wireless

From the examples discussed, it is evident that there is no end to the types of sensors or the range of applications each might be put to. The scope has increased immensely, thanks to the wireless generation of sensors. An application is no longer constrained by cumbersome wires, and it is possible to install a sensor practically anywhere.

Vivek Tyagi, country manager-sales, Freescale Semiconductor India explains, "Wireless has gained popularity over its wired counterpart primarily due to its vast applications and lowered cost of setup. Globally, wireless sensors are popularly used for military, medical and environmental applications since these are the most critical areas that need to be addressed. Apart from these, automotive, aerospace, manufacturing and consumer electronics sectors are also heavily dependent on these sensors."

A sensor hardly operates individually, although each single node is typically an autonomous instrument. Sensor nodes always act cooperatively with other sensors, as a network, with each node communicating with the others and/or with a central computer that acts as the 'leader.'

Depending on the setup, the sensor nodes might use short-range wireless technologies such as radio frequency identification (RFID) or Bluetooth to communicate with neighbouring sensors, or long-range communication technologies like Wi-Fi to communi-